

RIGA TECHNICAL UNIVERSITY

Faculty of Computer Science, Information Technology and Energy

Practical Assignment

In Course

“Knowledge management systems”

AuraDB Knowledge Graph Creation

Student name, surname:

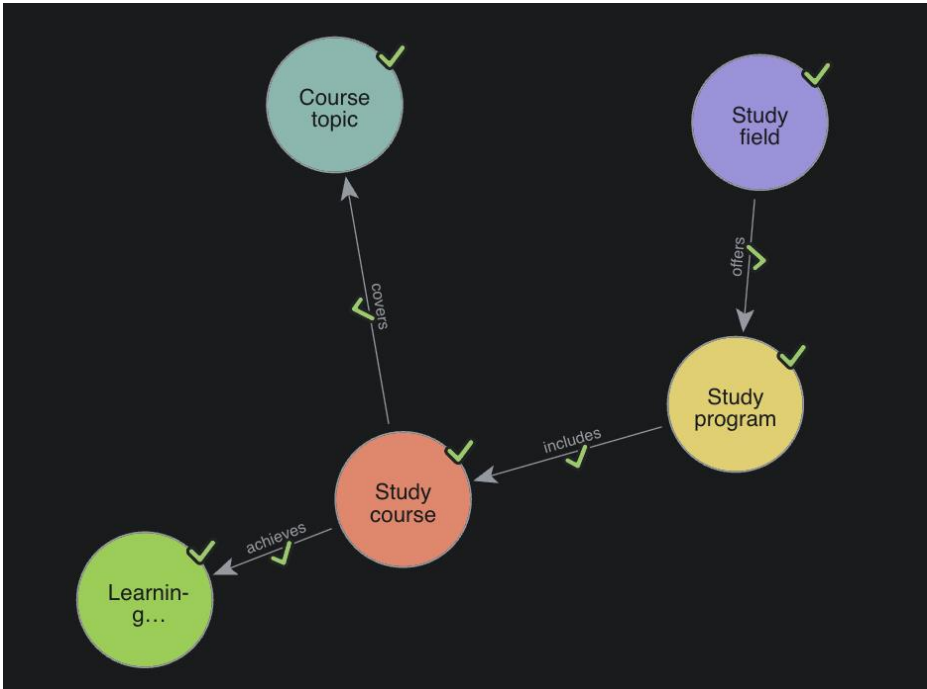
Krišjānis Dzenis

Student ID:

221RDB029

2026/2027

Data importing



Element	Schema
Node	Label ⓘ Study field Properties Constraints Indexes Properties + nodes_study_fields.csv Map from table ⌵ ☐ Name ☐ study_field_id ⓘ ☐ name ⓘ Type string string Column study_field_id name ID One or more properties that uniquely identify the node study_field_id × × ⌵

Relationship

Relationship type ⓘ

↔ Reverse direction

offers

Properties

Constraints

Indexes

Properties

+

program+study_field.csv

Map from table

🔍

☐

Name

Type

Column

☐

program_id

string

program_id

☐

study_field_id

string

study_field_id

Node ID mapping

Node

ID property ⓘ

ID column ⓘ

From

Study field

study_field_id

study_field_id

To

Study progr...

program_id

program_id

Need help?

ID

One or more properties that uniquely identify the relationship

Optional

Select...

Node

Label ⓘ

Study program

Properties

Constraints

Indexes

Properties

+

nodes_study_programs.csv

Map from table

☐

Name

Type

Column

☐

program_id ⓘ

string

program_id

☐

title ⓘ

string

title

☐

study_mode ⓘ

string

study_mode

☐

languages ⓘ

string

languages

☐

credit_points ⓘ

float

credit_points

☐

prerequisites ⓘ

string

prerequisites

☐

future_empl... ⓘ

string

future_employe...

☐

professional... ⓘ

string

professional_clas...

ID

One or more properties that uniquely identify the node

program_id

×

▼

Relationship

Relationship type ⓘ

↔ Reverse direction

Properties

Constraints

Indexes

Properties

+

Map from table

☐	Name	Type	Column
☐	creditPoints	float	credit_points
☐	sectionCode	string	section_code
☐	sectionName	string	section_name
☐	subsectionN...	string	subsection_name

Node ID mapping

	Node	ID property ⓘ	ID column ⓘ
From	Study progr...	program_id	program_id
To	Study course	course_id	course_id

Need help?

ID

One or more properties that uniquely identify the relationship

Optional

Select...

Node	Label ⓘ Study course Properties Constraints Indexes Properties + nodes_study_courses.csv Map from table ☐ Name Type Column ☐ course_id string course_id ☐ credit_points float credit_points ☐ languages string languages ☐ aims string aims ☐ title string title ID One or more properties that uniquely identify the node course_id ×
Relationship	Relationship type ⓘ ↔ Reverse direction covers Properties Constraints Indexes Properties + course+topic.csv Map from table ☐ Name Type Column ☐ course_id string course_id ☐ topic_id string topic_id Node ID mapping Node ID property ⓘ ID column ⓘ From Study course course_id course_id To Course topic topic_id topic_id Need help? ID One or more properties that uniquely identify the relationship Optional Select...

Node	Label ⓘ Course topic Properties Constraints Indexes Properties + nodes_course_topics.csv Map from table ☐ Name Type Column ☐ topic_id string topic_id ☐ course_id string course_id ☐ title string title ID One or more properties that uniquely identify the node topic_id X X
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Node

Label ⓘ

Learning outcome

Properties

Constraints

Indexes

Properties

+

nodes_learning_outcomes.csv

▼

Map from table

🔍

☐	Name	Type	Column
☐	outcome_id	string ▼	outcome_id ▼
☐	text	string ▼	text ▼
☐	scope	string ▼	scope ▼
☐	owner_id	string ▼	owner_id ▼

ID

One or more properties that uniquely identify the node

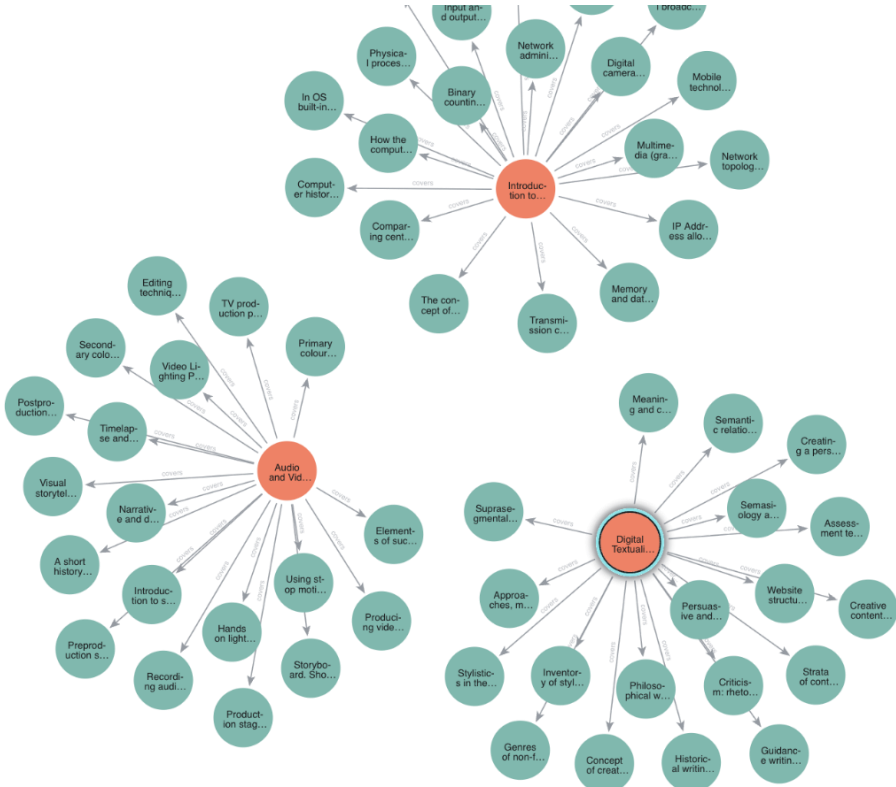
outcome_id

×

▼

Queries

Query #1	MATCH p=(:'Study field')-[:offers]->(:'Study program')-[:includes]->(:'Study course') RETURN p
Result	
Description	Displays the connection between “Study field” and “Study program” nodes and their courses
Query #2	MATCH (c:'Study course')-[:covers]->(t:'Course topic') WITH c, count(t) AS n ORDER BY n DESC LIMIT 3 MATCH (c)-[r:covers]->(t:'Course topic') RETURN c, r, t

Result	 The image displays three radial diagrams, each representing a central concept with its associated topics. The first diagram, 'Introduction to...', has a central red node connected to 15 green nodes including 'Input and output...', 'Network admin...', 'Digital camera...', 'Mobile technol...', 'Multimedia (gra...', 'Network topolog...', 'IP Addr...', 'Memory and dat...', 'Transmission c...', 'The concept of...', 'Comparing cent...', 'How the comput...', 'In OS built-in...', 'Physical proces...', and 'Binary countin...'. The second diagram, 'Audio and Vid...', has a central red node connected to 15 green nodes including 'Editing techniq...', 'TV production p...', 'Primary colour...', 'Video Li...', 'Postproduction...', 'Timelapse and...', 'Visual storytel...', 'Narrative and d...', 'A short history...', 'Introduction to s...', 'Preproduction s...', 'Recording audi...', 'Production stag...', 'Using stop moti...', 'Storybo...', 'Producing vide...', and 'Element-s of suc...'. The third diagram, 'Digital Textual...', has a central red node connected to 15 green nodes including 'Meaning and c...', 'Semanti-c relatio...', 'Creatin-g a pers...', 'Assess-ment le...', 'Creative content...', 'Website structu...', 'Persuas-ive and...', 'Critica-m. rieto...', 'Strata of cont...', 'Guidanc-e writin...', 'Historic-al writin...', 'Concept of creat...', 'Philoso-phical w...', 'Inventor-y of styl...', 'Stylistic-s in the...', 'Approach-es, m...', 'Suprase-gmental...', and 'Meaning and c...'. Each diagram uses arrows to show the relationship between the central concept and its topics.
Description	Returns the first 3 courses that have the most covered topics
Query #3	<i>MATCH</i> (c: `Study course` {course_id: "DE0745"}) <i>OPTIONAL MATCH</i> (c)-[r1:covers]->(t: `Course topic`) <i>OPTIONAL MATCH</i> (c)-[r2:achieves]->(o: `Learning outcome`) <i>RETURN</i> c, r1, t, r2, o

Result	
Description	Returns specific course “DE0745” covered topics and learning outcomes
Query #4	MATCH (o: `Learning outcome`) WHERE NOT (()-[:achieves]->(o)) RETURN o
Result	
Description	Returns learning outcomes that are not linked to any course. Query results show that 30 learning outcomes are not linked with any course, indicating issue in data quality.

Query #5	<i>MATCH</i> (p: `Study program`)-[:includes]->(c: `Study course`) <i>WITH</i> c, count(p) AS k <i>WHERE</i> k > 1 <i>MATCH</i> (p2: `Study program`)-[r:includes]->(c) <i>RETURN</i> p2, r, c
Result	<pre> graph TD DH((Digital Humanit...)) C((Commu-nication...)) KM((Knowle-dge Ma...)) PM((Portfoli-o Mana...)) EB((e-Busin-ess...)) B((Busines-s...)) DH -- includes --> C DH -- includes --> KM DH -- includes --> PM DH -- includes --> EB B -- includes --> C B -- includes --> KM B -- includes --> PM B -- includes --> EB </pre>
Description	Finds courses that overlap between study programs. Result show that Business Informatics and Digital Humanities programme students will attend 4 courses together